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 Math 454 – Assignment 5

Problem 1. We can prove integrability and the inequality at once. Note that as f and g are integrable, so are $|f|$ and $|g|$ (by definition). That is,

$$\int_{\mathbb{R}} |f| < \infty, \quad \int_{\mathbb{R}} |g| < \infty$$

To these we will apply *Tonelli's Theorem*. To show integrability of $(f \star g)$ we will show $|f \star g|$ is integrable. We argue as follows:

$$\begin{aligned} \int_{\mathbb{R}} |f \star g|(x) dx &= \int_{\mathbb{R}} \left| \left(\int_{\mathbb{R}} f(x-y)g(y) dy \right) \right| dx \leq \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(x-y)| |g(y)| dy \right) dx \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(x-y)| |g(y)| dx \right) dy = \int_{\mathbb{R}} |g(y)| \left(\int_{\mathbb{R}} |f(x-y)| dx \right) dy \end{aligned}$$

By translation invariance, we see this last line is equivalent to

$$\int_{\mathbb{R}} |g(y)| \left(\int_{\mathbb{R}} |f(x)| dx \right) dy = \int_{\mathbb{R}} |f(x)| dx \int_{\mathbb{R}} |g(y)| dy = \|f\|_{L^1(\mathbb{R})} \cdot \|g\|_{L^1(\mathbb{R})} < \infty$$

as both these functions are integrable. Putting all this together yields

$$\|f \star g\|_{L^1(\mathbb{R})} \leq \|f\|_{L^1(\mathbb{R})} \cdot \|g\|_{L^1(\mathbb{R})} \quad (1)$$

as was required. Moreover, an identical argument applies for $\mathbb{R}^n$.

Problem 2. By problem 1 the convolution is integrable. Writing $f = f^+ - f^-$ we see that Tonelli applies. We may directly express the transform of $f \star g$ as

$$\widehat{f \star g}(x) = \int_{\mathbb{R}} (f \star g)(t) e^{-2\pi i x \cdot t} dt = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(t-y)g(y) dy \right) e^{-2\pi i x \cdot t} dt \quad (2)$$

We may now directly apply Tonelli's Theorem to see:

$$\begin{aligned} \widehat{f \star g}(x) &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(t-y)g(y) dy \right) e^{-2\pi i x \cdot t} dt \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} f(t-y)g(y) e^{-2\pi i x \cdot t} dy dt = \int_{\mathbb{R}} \int_{\mathbb{R}} f(t-y)g(y) e^{-2\pi i x \cdot t} \cdot \frac{e^{-2\pi i x y}}{e^{-2\pi i x y}} dy dt \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} f(t-y) e^{-2\pi i (x-y) \cdot t} g(y) e^{-2\pi i x y} \cdot dy dt = \int_{\mathbb{R}} \int_{\mathbb{R}} f(t-y) e^{-2\pi i (x-y) \cdot t} g(y) e^{-2\pi i x y} \cdot dt dy \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(t-y) e^{-2\pi i (x-y) \cdot t} \cdot dt \right) g(y) e^{-2\pi i x y} dy = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(t) e^{-2\pi i x \cdot t} \cdot dt \right) g(y) e^{-2\pi i x y} dy \\ &= \int_{\mathbb{R}} g(y) e^{-2\pi i x y} dy \cdot \int_{\mathbb{R}} f(t) e^{-2\pi i x \cdot t} \cdot dt = \widehat{f}(x) \cdot \widehat{g}(x) \end{aligned}$$

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We now show there is no integrable I so that $f \star I = f$ for all f integrable. Of course, we argue by contradiction. Suppose such I exists. Then, as $f \star I = f$ for all f one has by the proceeding argument

$$\widehat{f}(x)\widehat{I}(x) = \widehat{f \star I}(x) = \widehat{f}(x)$$

since we may take any f , it must follow that $\widehat{I}(x) = 1$ for all x (this is where we need the "for all f "). As such, taking $x \rightarrow \infty$ one has:

$$\lim_{x \rightarrow \infty} \widehat{I}(x) = \lim_{x \rightarrow \infty} \int_{\mathbb{R}} I(\sigma) e^{-2\pi i x \sigma} d\sigma = 1 \quad (3)$$

Let us now recall the Borel-Cantelli lemma we proved in the previous assignment (Problem 4). This state that if f is integrable then

$$\int_{\mathbb{R}} f(x) e^{2\pi i n x} dx \xrightarrow{n \rightarrow \infty} 0 \quad (\mathcal{B} - \mathcal{C})$$

However, a verbatim argument will show that for such f

$$\int_{\mathbb{R}} f(x) e^{-2\pi i n x} dx \xrightarrow{n \rightarrow \infty} 0$$

Clearly, (3) contradicts this. Thus, no such I exists.

Problem 3. First we note that showing the inequality is sufficient to prove $f^\star$ is not-integrable. Indeed, if the inequality holds then

$$\int_{\mathbb{R}} f^\star(x) dx \geq \int_1^\infty f^\star(x) dx \geq \int_1^\infty \frac{c}{x} dx = \infty$$

by the integral test. With this out of the way, let us now turn to the main proof of the problem. Since f is not equivalent to 0, we may find a ball, say, $B(x_0, \rho)$, over which

$$\int_{B(x_0, \rho)} |f| = \delta > 0, \quad \rho > 0 \text{ small}$$

Now let $x \in \mathbb{R}$ be given with $|x| \geq 1$. Consider now the interval $B(0, |x| + \rho + |x_0|)$. We now claim that $B(x_0, \rho) \subseteq B(0, |x| + |x_0| + \rho)$. This is clear writing

$$|y| = |y - x_0 + x_0| \leq \rho + |x_0| + |x|, \quad y \in B(x_0, \rho)$$

Now, as ρ and x_0 are fixed for this f , we may take A so large that $|x| + |x_0| + \rho \leq A|x|$ whenever $|x| \geq 1$. Thus,

$$f^\star(x) = \sup_{B \ni x} \frac{1}{\lambda(B)} \int_B |f| \geq \frac{1}{\lambda(B(0, \rho + |x_0| + |x|))} \int_{B(0, \rho + |x_0| + |x|)} |f| \quad (4)$$

$$\geq \frac{A_1}{|x|} \int_{B(x_0, \rho)} |f| = \frac{A_1 \delta}{|x|} > \frac{c}{|x|} > 0 \quad (5)$$

taking $c = \frac{A_1 \delta}{2}$ for instance.

Problem 4. Since $\lambda(E) = 0$, taking $\epsilon := \frac{1}{2^n}$ for $n \in \mathbb{N}$, we may pick an open set $\mathcal{O}_n \supseteq E$ with $\lambda(\mathcal{O}_n) \leq \frac{1}{2^n}$. Let $\{\mathcal{O}_n\}_{n \in \mathbb{N}}$ be the resulting sequence of sets. Define now the function

$$f(x) := \sum_{n \in \mathbb{N}} \mathbb{1}_{\mathcal{O}_n}(x) \quad (f)$$

f is clearly non negative and is integrable, for

$$\int_{\mathbb{R}} f = \sum_{n \in \mathbb{N}} \int_{\mathbb{R}} \mathbb{1}_{\mathcal{O}_n}(x) dx \leq \sum_{n \in \mathbb{N}} \frac{1}{2^n} = 1 < \infty$$

Fix $x \in E$ arbitrarily and observe that $x \in \mathcal{O}_n$ for all $n \in \mathbb{N}$ by construction of such covers. We first need to show that x is a density point of all open sets containing it. Indeed, if $x \in O$, O open, then eventually, all small enough balls containing x eventually are subsets of the covering set O , by definition of an open set. Thus, for all small enough $B_\epsilon \ni x$ we have $B_\epsilon \cap O = B_\epsilon$ so that

$$\lim_{\lambda(B_\epsilon) \rightarrow 0} \frac{\lambda(O \cap B_\epsilon)}{\lambda(B_\epsilon)} = \lim_{\lambda(B_\epsilon) \rightarrow 0} \frac{\lambda(B_\epsilon)}{\lambda(B_\epsilon)} = 1$$

Now define an arbitrary sequence $(B_k)_{k \in \mathbb{N}}$ of open intervals about x with $\lambda(B_k) \xrightarrow{k \rightarrow \infty} 0$. For any such index k we will consider

$$\begin{aligned} \frac{1}{\lambda(B_k)} \int_{B_k} f &= \frac{1}{\lambda(B_k)} \sum_{n \in \mathbb{N}} \int_{B_k} \mathbb{1}_{\mathcal{O}_n}(x) dx = \frac{1}{\lambda(B_k)} \sum_{n \in \mathbb{N}} \int_{\mathbb{R}} \mathbb{1}_{\mathcal{O}_n \cap B_k}(x) dx \\ &= \sum_{n \in \mathbb{N}} \frac{\lambda(B_k \cap \mathcal{O}_n)}{\lambda(B_k)} \end{aligned}$$

We need to study the behaviour of the summation as $k \rightarrow \infty$ (i.e $\lambda(B_k) \rightarrow 0$). Now, since x is a point of Lebesgue density (of any $\mathcal{O}_n$) we have by definition

$$\lim_{k \rightarrow \infty} \frac{\lambda(B_k \cap \mathcal{O}_n)}{\lambda(B_k)} = 1, \quad \forall n \in \mathbb{N} \quad (6)$$

It is then clear that the following series is divergent

$$\sum_{n \in \mathbb{N}} \lim_{k \rightarrow \infty} \frac{\lambda(B_k \cap \mathcal{O}_n)}{\lambda(B_k)} = \infty$$

Then, in particular

$$\sum_{n \in \mathbb{N}} \liminf_{k \rightarrow \infty} \frac{\lambda(B_k \cap \mathcal{O}_n)}{\lambda(B_k)}$$

whence we derive

$$\begin{aligned}
 \infty &= \sum_{n \in \mathbb{N}} \liminf_{k \rightarrow \infty} \frac{\lambda(B_k \cap \mathcal{O}_n)}{\lambda(B_k)} \leq \liminf_{k \rightarrow \infty} \sum_{n \in \mathbb{N}} \frac{\lambda(B_k \cap \mathcal{O}_n)}{\lambda(B_k)} \\
 &= \liminf_{k \rightarrow \infty} \frac{1}{\lambda(B_k)} \sum_{n \in \mathbb{N}} \int_{\mathbb{R}} \mathbb{1}_{\mathcal{O}_n \cap B_k}(x) dx = \liminf_{k \rightarrow \infty} \frac{1}{\lambda(B_k)} \int_{B_k} \sum_{n \in \mathbb{N}} \mathbb{1}_{\mathcal{O}_n}(x) dx \\
 &= \liminf_{k \rightarrow \infty} \frac{1}{\lambda(B_k)} \int_{B_k} f(x) dx
 \end{aligned}$$

Since the sequence of balls was taken arbitrarily we recover

$$\liminf_{\lambda(B) \rightarrow 0} \frac{1}{\lambda(B)} \int_B f(x) dx \tag{7}$$

Finally, as x was arbitrary the desired result follows.

Problem 5. It follows from hypothesis on E that $\lambda(E) \leq 1$. It only remains to show that $\lambda(E) \geq 1$ whence it will follow that $\lambda(E) = 1$. By Corollary 1.5 in the textbook one has that almost every point of E^c is not a point of Lebesgue density. More precisely, there exists a set $X \subset E^c$ of measure 0 so that

$$x \in E^c \setminus X \implies x \text{ is not a point of Lebesgue density}$$

It turns out that for almost all $x \notin E$ the limit

$$\lim_{\substack{\lambda(B) \rightarrow 0 \\ B \ni x}} \frac{\lambda(B \cap E)}{\lambda(B)} = 0 \tag{8}$$

Indeed, this follows from the *Lebesgue Differentiation Theorem* taking $f = \mathbb{1}_E$:

$$\mathbb{1}_E(x) = \lim_{\substack{\lambda(B) \rightarrow 0 \\ B \ni x}} \frac{1}{\lambda(B)} \int_B \mathbb{1}(y) dy, \quad a.e$$

In particular,

$$0 = \lim_{\substack{\lambda(B) \rightarrow 0 \\ B \ni x}} \frac{1}{\lambda(B)} \int_B \mathbb{1}(y) dy = \lim_{\substack{\lambda(B) \rightarrow 0 \\ B \ni x}} \frac{\lambda(B \cap E)}{\lambda(B)} \tag{9}$$

for almost all $x \in E^c$. Therefore, there exists a measurable set $X \subset \mathbb{R}$ with $\lambda(X) = 0$ so that

$$\lim_{\substack{\lambda(B) \rightarrow 0 \\ B \ni x}} \frac{\lambda(B \cap E)}{\lambda(B)} = 0, \quad \forall x \in E^c \setminus X$$

This is true, in particular for the subset $[0, 1] \cap (E^c \setminus X)$. For simplicity we now consider the interior of $[0, 1]$. Pick any $x \in (0, 1)$. Then there exists infinitely many small intervals, say I_ϵ , containing x such that $I_\epsilon \subseteq (0, 1) \subset [0, 1]$. For each of these intervals,

$$\frac{\lambda(I_\epsilon \cap E)}{\lambda(I_\epsilon)} \geq \alpha > 0$$

Thus, for all $x \in (0, 1)$ one has

$$\lim_{\substack{\lambda(B) \rightarrow 0 \\ B \ni x}} \frac{\lambda(B \cap E)}{\lambda(B)} \neq 0$$

whether or not the limit exists. That is,

$$\begin{aligned} x \notin [0, 1] \cap (E^c \setminus X) &= [0, 1] \cap (E^c \cap X^c) \\ &= [0, 1] \cap (E \cup X)^c = [0, 1] \setminus (E \cup X) \end{aligned}$$

for otherwise the limit in question would indeed be 0. Consequently, we have that $x \in E \cup X$ and hence $(0, 1) \subseteq E \cup X$. Here is where we use that X has measure 0. In this case,

$$1 = \lambda((0, 1)) \leq \lambda(E) + \lambda(X) = \lambda(E)$$

which gives us the reverse inequality. Therefore $\lambda(E) = 1$.